

Enterprise AI Architect Learning Path v3 – Phase Guide 3

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Production RAG & Evaluation

Purpose of This Phase

Phase 2 focused on building a working Retrieval-Augmented Generation (RAG) platform.

Phase 3 focuses on transforming that platform into an enterprise-grade retrieval system.

The goal is to move beyond:

"The system usually returns good answers."

to:

"The system consistently returns measurable, explainable, and trusted answers."

In enterprise environments, retrieval quality cannot be judged by intuition alone.

RAG systems must be:

- Evaluated
- Measured
- Tested
- Monitored
- Improved systematically

This phase introduces evaluation as a first-class architectural discipline.

Why This Phase Matters

Many RAG systems fail because their teams only evaluate:

- Whether the answer sounds reasonable

instead of evaluating:

- Whether the correct information was retrieved
- Whether citations are accurate
- Whether answers are properly grounded
- Whether performance remains consistent over time

Architects need objective measures of quality.

This phase establishes the foundation for:

- AI quality programs
 - Evaluation pipelines
 - Retrieval governance
 - Production AI operations
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Phase Objectives

Upon completion of this phase, you should be able to:

- Design retrieval evaluation programs
 - Measure retrieval quality
 - Implement citation contracts
 - Build hybrid retrieval solutions
 - Create golden datasets
 - Identify retrieval failure modes
 - Improve retrieval performance systematically
 - Separate retrieval quality from generation quality
 - Build repeatable evaluation workflows
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Architect Mindset

A common mistake in AI projects is blaming the model for failures that originate in retrieval.

Architects learn to diagnose systems by asking:

Did retrieval find the correct information?

If not:

- Search quality problem

Did retrieval find the correct information but the model ignored it?

If yes:

- Prompting or orchestration problem

Did retrieval and prompting both succeed?

If yes:

- Output generation problem

The architecture must enable these distinctions.

Primary Resources

Required

R1

AI Engineering

Focus Areas:

- Retrieval failure modes
 - RAG evaluation
 - Prompt versioning
 - Grounding strategies
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R2

Designing Machine Learning Systems

Focus Areas:

- Metrics
 - Evaluation
 - Monitoring
 - Production quality processes
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R18

Generative AI with LangChain

Focus Areas:

- Advanced RAG patterns
 - Evaluation workflows
 - Production testing
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R5

Azure AI Search Documentation

Focus Areas:

- Hybrid search
 - Filtering
 - Ranking improvements
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R7

Azure AI Search Samples

Focus Areas:

- Search optimization
 - Advanced retrieval techniques
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Core Technical Skills

1. Hybrid Search

You implemented basic retrieval in Phase 2.

Now refine retrieval quality using:

Keyword Search

Strengths:

- Exact matches
 - Names
 - Rule numbers
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Vector Search

Strengths:

- Semantic similarity
 - Meaning-based retrieval
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Hybrid Search

Strengths:

- Combines semantic relevance
- Retains keyword precision

Enterprise systems often favor hybrid approaches because they balance accuracy and usability.

2. Metadata Filtering

Learn how metadata can improve retrieval.

Examples:

Rule System

Campaign Setting

Source Book

Adventure Type

Publisher

Version

A good filter often improves retrieval more than changing the model.

3. Citation Contracts

The system should define:

Required Behavior

Every answer:

- Includes sources
 - Includes document references
 - Distinguishes fact from inference
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Failure Behavior

When support cannot be found:

The system should refuse to fabricate an answer.

4. Retrieval Evaluation

Measure:

Did the correct content appear?

Was it ranked appropriately?

Were sources correct?

Was grounding maintained?

These become measurable quality metrics.

5. Golden Question Sets

Build a collection of known questions.

These should represent:

- Simple facts
- Complex rules
- Ambiguous scenarios
- Edge cases

Golden Questions become long-term regression tests for the retrieval system.

Phase Project

RPG Catalog RAG v1

This phase upgrades the RAG platform from a functional prototype into an evaluated retrieval system.

Functional Requirements

Improved Retrieval

Support:

- Hybrid Search
 - Metadata Filters
 - Search Tuning
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Improved Citations

Every response should:

- Reference sources
 - Expose retrieval origins
 - Support verification
-

Evaluation Framework

Capability to:

- Execute standardized test questions
 - Analyze failures
 - Measure improvements
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Search Quality Improvements

Continuously improve:

- Precision
- Relevance
- Grounding

without changing core architecture unnecessarily.

Golden Question Program

Objective

Create a retrieval quality benchmark.

Suggested Categories

Rules Questions

Example:

What are the initiative rules?

Lore Questions

Example:

Who founded the kingdom?

Cross-Document Questions

Example:

How does Character X connect to Event Y?

Negative Questions

Example:

Tell me about something not present in the source materials.

“

Expected behavior:

Refusal to fabricate.

Target Size

Minimum:

50 Questions

Stretch Goal:

100 Questions

organized by category.

Architecture Deliverables

ADR-007

Hybrid Retrieval Strategy

Document:

- Why hybrid search was selected
- Tradeoffs considered
- Ranking strategy

ADR-008

Citation Contract

Document:

- Citation expectations
- Refusal rules
- Grounding requirements

ADR-009

Evaluation Strategy

Document:

- Golden Question design
- Measurement approach
- Testing process

Portfolio Artifacts

Evaluation Report

Include:

- Test execution results
- Successes
- Failures
- Lessons learned

Retrieval Metrics Dashboard

Track:

- Question categories
- Retrieval quality
- Evaluation history

Updated Architecture Diagrams

Reflect:

- Evaluation layer
 - Retrieval flow
 - Search ranking process
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Retrieval Decision Log

Document:

- Search changes
 - Chunking changes
 - Ranking experiments
 - Metadata experiments
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Suggested Sprint Progression

Sprint 1 — Retrieval Assessment

Goals:

- Review current platform
- Identify weaknesses
- Create improvement backlog

Done When:

- Evaluation plan exists
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Sprint 2 — Hybrid Search

Goals:

- Implement hybrid retrieval
- Compare search approaches

Done When:

- Search quality improved
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Sprint 3 — Metadata Strategy

Goals:

- Expand metadata usage
- Add filtering

Done When:

- Filters operational
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Sprint 4 — Citation Improvements

Goals:

- Improve grounding
- Improve source traceability

Done When:

- Citation contract implemented
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Sprint 5 — Golden Dataset

Goals:

- Build question set
- Categorize coverage

Done When:

- Golden questions completed
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Sprint 6 — Evaluation Pipeline

Goals:

- Execute tests
- Measure quality
- Identify failures

Done When:

- Evaluation report produced
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Sprint 7 — Optimization

Goals:

- Improve retrieval
- Improve ranking
- Reduce failure modes

Done When:

- RPG Catalog RAG v1 completed
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Common Pitfalls

Pitfall 1

Evaluating Answers Instead of Retrieval

The answer can sound good while being completely unsupported.

Always evaluate retrieval independently.

Pitfall 2

Changing Models Instead of Retrieval

Poor retrieval quality cannot be fixed by repeatedly changing models.

Investigate:

- Chunking
- Metadata
- Ranking
- Filters

first.

Pitfall 3

No Baseline

Without golden questions:

You cannot tell whether changes are improvements or regressions.

Pitfall 4

Ignoring Failure Cases

The most valuable questions are often:

Why did retrieval fail?

rather than:

Why did it succeed?

Enterprise Translation Layer

The evaluation architecture developed here applies directly to enterprise systems.

	≡ RPG Catalog Use Case	≡ Enterprise Equivalent
1	Rule Retrieval	Policy Retrieval
2	Lore Search	Knowledge Management
3	Adventure Content	Technical Documentation
4	Golden Questions	Compliance Test Suites
5	Citation Validation	Regulatory Traceability

The evaluation methodology remains the same regardless of content domain.

Definition of Done

Phase 3 is complete when:

- Hybrid search is operational
 - Metadata filtering exists
 - Citation contract is implemented
 - Golden Question set exists
 - Evaluation framework is operational
 - Retrieval failures are documented
 - Evaluation report is completed
 - ADRs are documented
 - Architecture diagrams are updated
 - RPG Catalog RAG v1 is demonstrably more reliable than v0.5
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End-of-Phase Self-Assessment

You are ready for Phase 4 when you can confidently answer:

- Can I objectively evaluate retrieval quality?
- Can I distinguish retrieval failures from generation failures?
- Can I explain why a search result was returned?
- Can I improve retrieval without changing models?
- Can I build and maintain an evaluation dataset?

- Can I justify the retrieval strategy I've chosen?

If all answers are "yes," proceed to **Phase 4 – Agent Orchestration**.